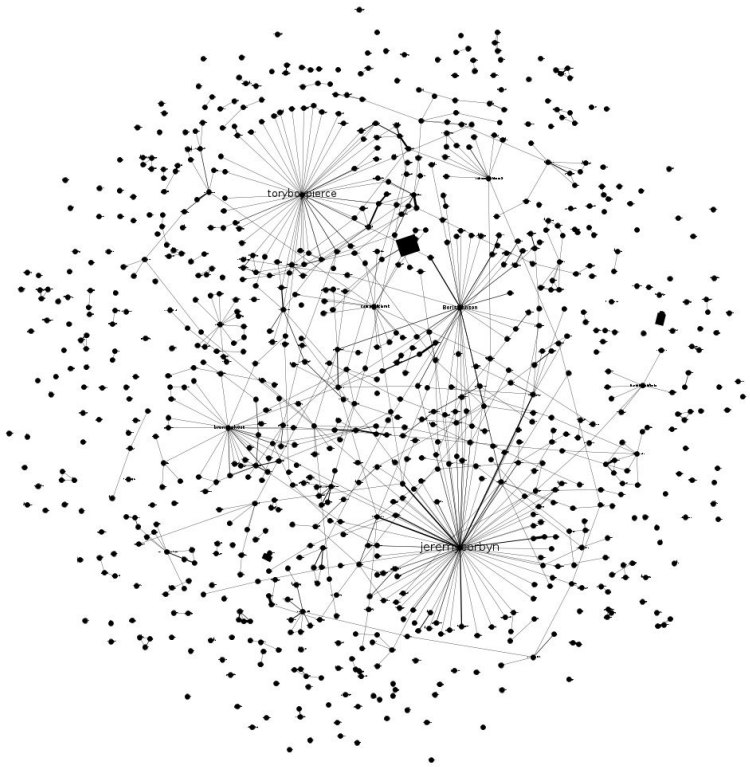
General election 2019, Twitter Network and Sentiment analysis

Overview

In this report I will describe how I used Python and the program Gephi[1] to scrape and explore twitter networks relating to the General election that took place in the UK in late 2019. By scraping Twitter data with Python, I used Gephi to look at the features of the related networks, specifically focusing on the leaders of the two main parties and their related party accounts. I then used the Natural Language Toolkit library (NLTK) to analyse sentiments of the tweets and plotted histograms of these aiming to see the spread and variance of sentiment across the party lines.



My main research enquiry was regarding what patterns and phenomena can be observed in the twitter networks surrounding political figures and entities. Furthermore to investigate what

[1] Bastian M., Heymann S., Jacomy M. (2009). ***Gephi: an open source software for exploring and manipulating networks.*** International AAAI Conference on Weblogs and Social Media.

aspects of the political landscape, as a whole, are reflected in the *twittersphere*, particularly whether one can observe party structures and campaign strategies.

Initial Gephi network showing a query-search for “@jeremycorbyn @borisjohnson”

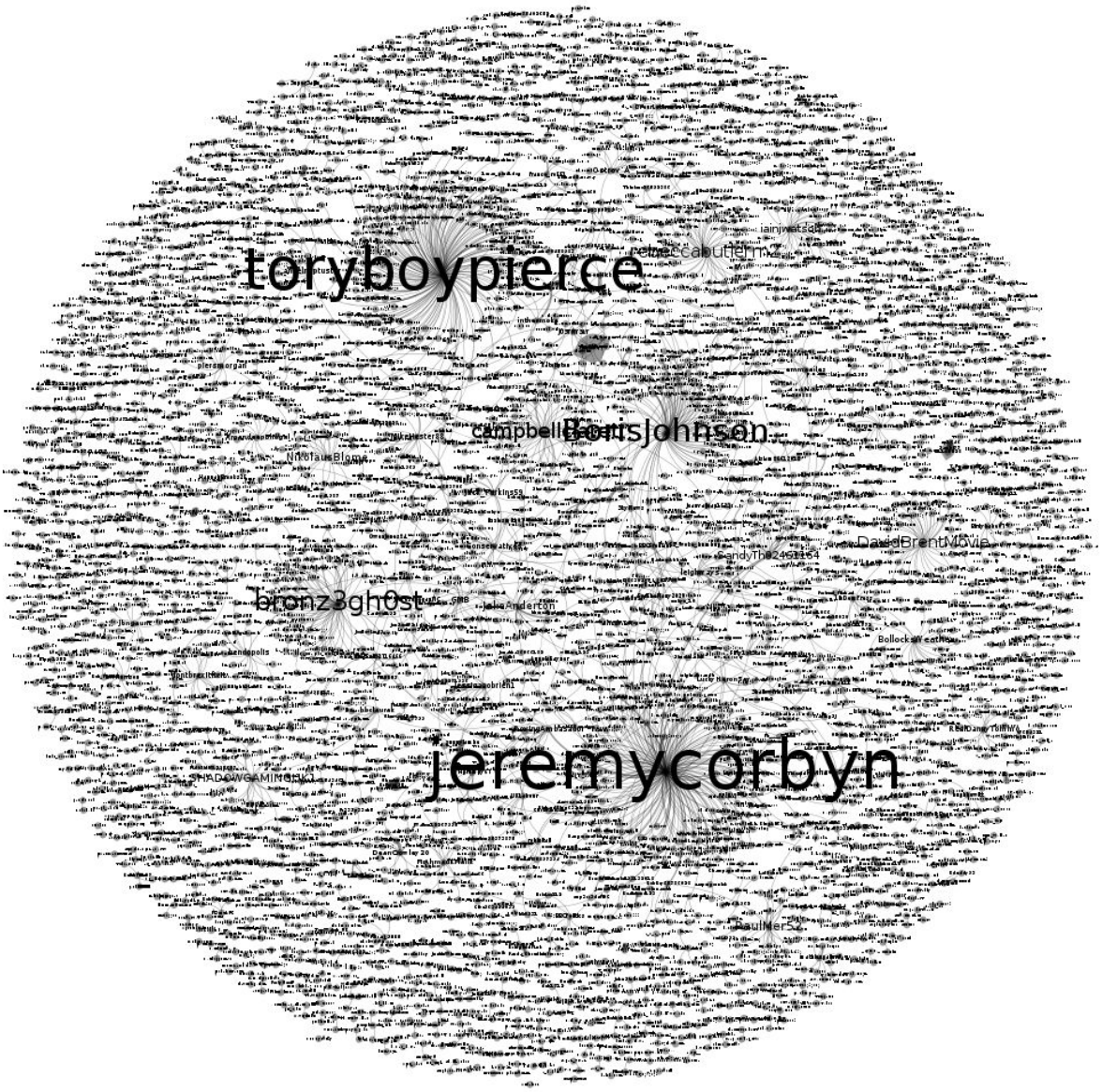
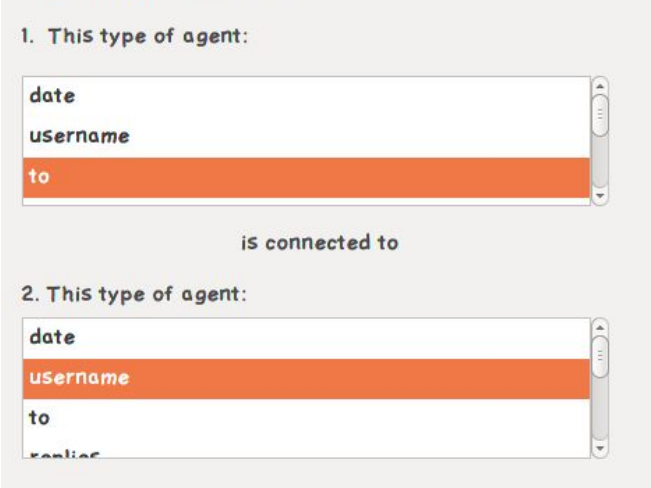


Fig. 1

```
GetOldTweets3 --querysearch @borisjohnson @jeremycorbyn --since 2019-12-01
--until 2019-12-13 --maxtweets 7100 --output jeris.csv
```

Above is the command I used to scrape the tweets for the network above that. I used the GetOldTweets3 Python package, as the twitter API has a 7-day search limit; it allowed me to scrape tweets from December 2019 while in January 2020. A time frame from the 1st December 2019 up to the 13th December 2019 (the day of the election result) was chosen to cover the 12 days before and including election. I chose this to cover a period running up to election day itself; the final flurry of activity where, I assume, political narratives are generally cemented and finalised. In this state the network structure relays the final movements, of both sides, social media efforts, also serving as a standard time window with which to gain observations and compare results. The "*Convert Excel and csv files to networks"* Gephi plugin was also used to convert my .csv outputs from GetOldTweets, into network files. This was a crucial part of my network construction pipeline as it was in this plugin that I made decisions about the structure of the network. While importing I selected what connections would be made in the network; in the below screenshot the "*to"* (who the tweet was tweeted to using @) column is connected to the "*username"* column. This created a network whereby a *username* is connected to other users which were mentioned by that user in each tweet; thus creating a map of who is interacting with who in the network (through a method of **replies** or **mentions**) .



This created some interesting structures and illustrates links between different parties and journalists and other popular twitter accounts. In the above graph (Fig. 1) an immediate observation is the relative size of Johnson and Corbyn's immediate clusters. There is talk of a left-wing skew [2] on twitter and the presence of ideological echo chambers (this may be observed here, certainly in Gephi, it is clear when zooming in and exploring connections that there are relatively few links between politically opposed individuals). The other large presence of Daily Mail journalist @toryboypierce shows an equally strong right-wing twitter player.

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2 https://www.telegraph.co.uk/technology/twitter/11524724/Twitter-is-an-unrepresentative-urban-liberal-dreamland.html, https://www.theguardian.com/technology/2012/aug/22/twitter-online-echo-chanber-leftwing

Separate @borisjohnson and @jeremycorbyn networks

```
GetOldTweets3 --query-search “@jeremycorbyn” --since 2019-12-01 --until
2019-12-13 --maxtweets 7100 --output borisjohnson.csv
```

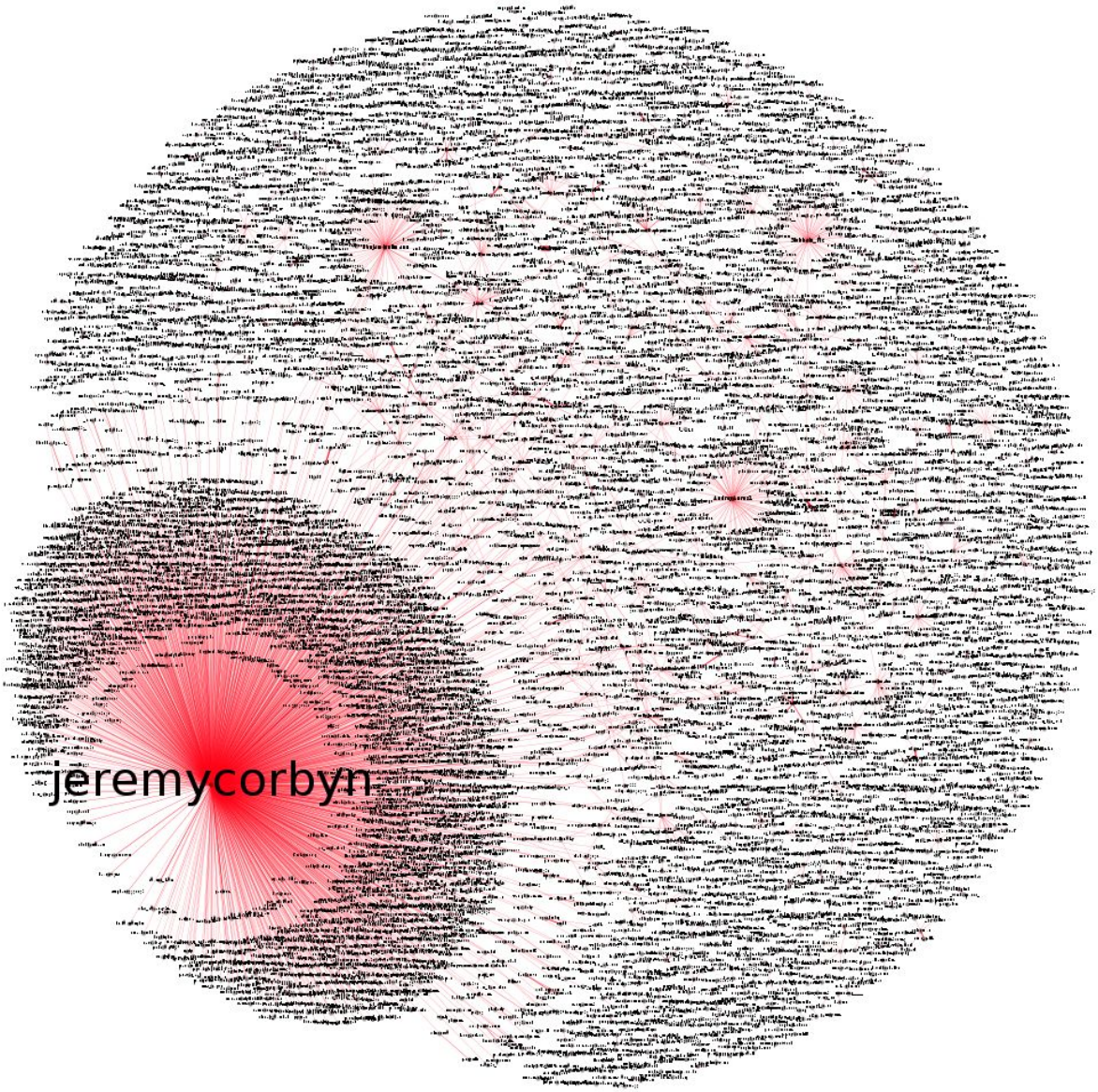


Fig. 2

Above (Fig. 2) is an initial network of a query-search of @jeremycorbyn (Jeremy Corbyn’s personal twitter account) which shows a characteristic shape also seen in @borisjohnsons equivalent network.

```
GetOldTweets3 --query-search “@borisjohnson” --since 2019-12-01 --until
2019-12-13 --maxtweets 7100 --output borisjohnson.csv
```

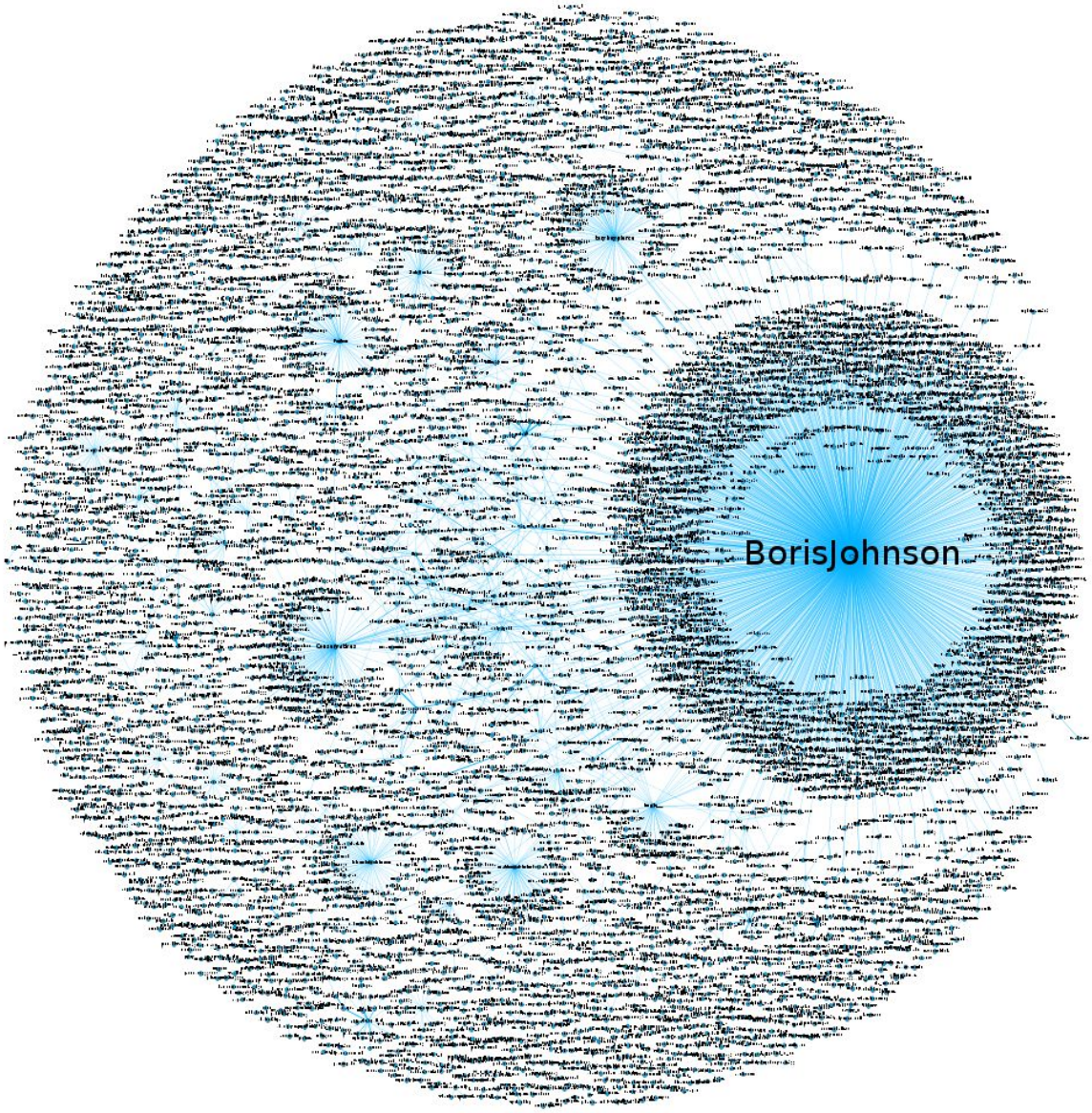
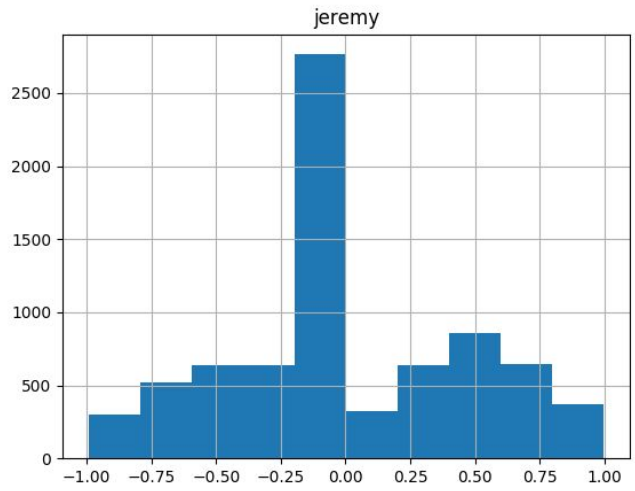
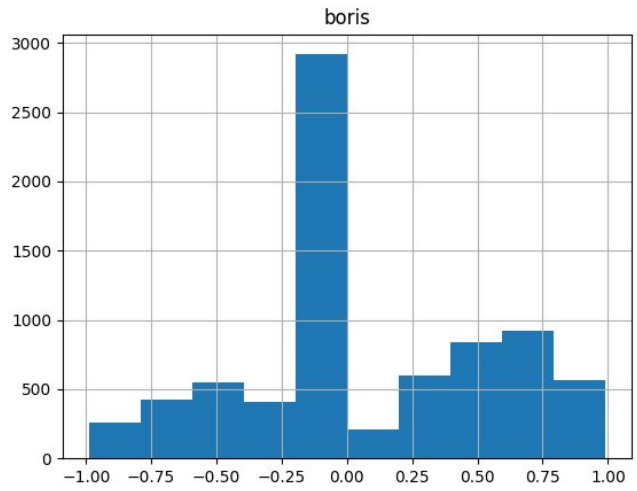


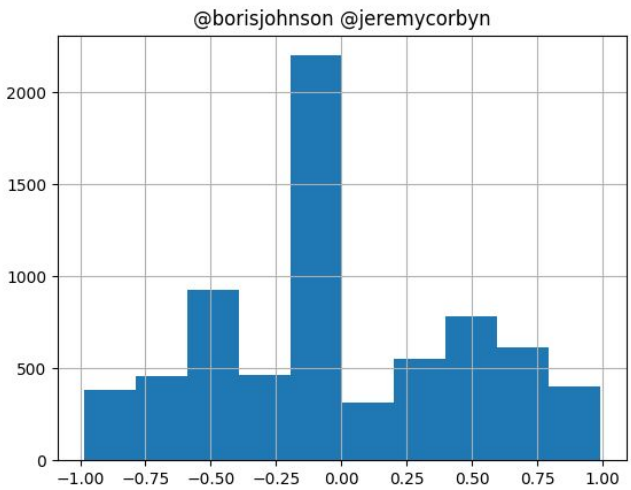
Fig.3

In Johnson’s network (Fig. 3) there does appear to be more larger structures dispersed around the network than in Corbyns. This may be indicative of a winning Conservative social media strategy or an indirect result of an overall stronger campaign across the board. Gephi’s *Average Degree* statistic is also higher for Johnson’s graph at 1.77 while Corbyns is lower at 1.68 which could suggest an overall more *social* network.

Sentiment analysis







Above you can see the histograms of compound sentiments (a scale of positive and negative sentiments, -1 being entirely negative, +1 being entirely positive) for the query-search of Johnson, Corbyn and the for them both. Negative values on the x axis represent correspond to negative sentiments while positive values to positive sentiments. Interestingly both have the same structure. This spike just left of zero (most likely neutral or unknown sentiments) could be from the use of pictures and videos which can't be analysed by the NLTK. There does appear to be a possible negative skew for Corbyn's histogram, with more negative values going above the 500 frequency line. I should reiterate that none of tweets in the *jeremy* and *boris* histograms immediately below are their own tweets, this is their surrounding networks; people who tweeted

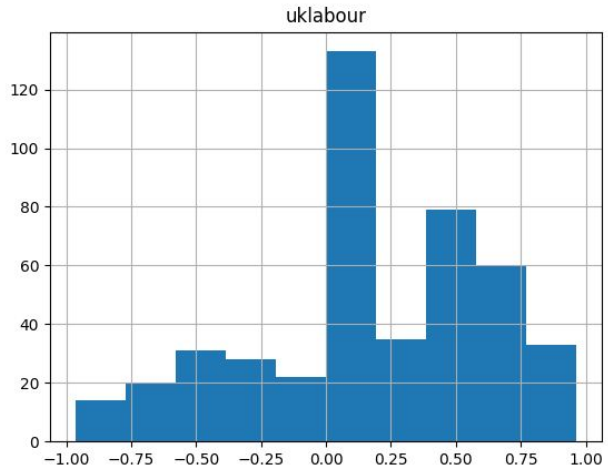
@ them. This may well just be an illustration of a generally more negative attitude towards Corbyn (an obvious outcome, given the election result)

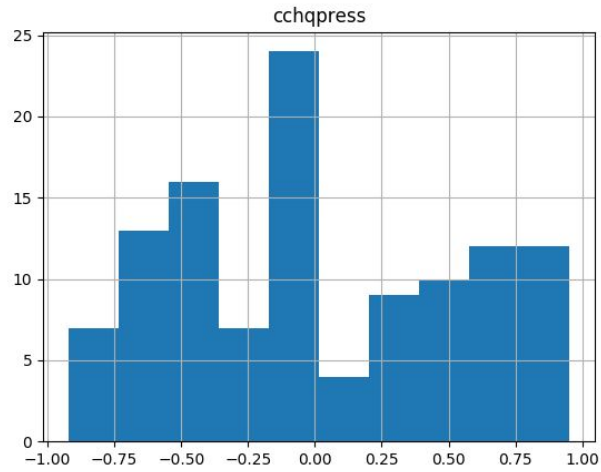
I went on to scrape the tweets belonging to @UKLabour and @CCHQPress (see below) during the same time interval and found gained a more pronounced difference (the commands are shown below). This time these tweets are ones that the two accounts tweeted themselves. The Conservative press account as a clear negative skew whereas the Labour Party has a definite positive skew. It is worth recognising that @CCHQPress is not the Conservative Party's main account has relatively little activity, although it could illustrate a wider trend.

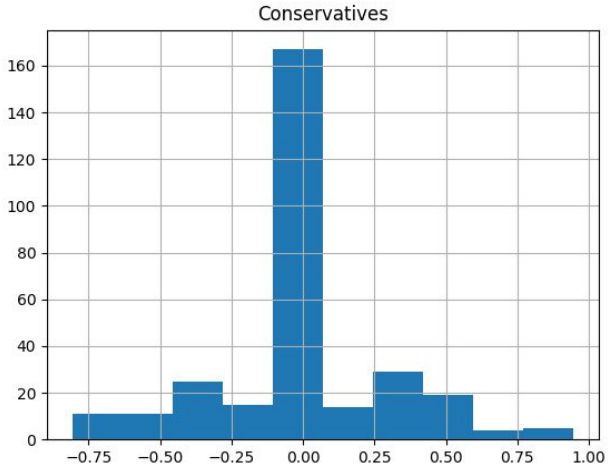
```
GetOldTweets3 --username "UKLabour" --since 2019-12-01 --until 2019-12-13
--output uklabour.csv

GetOldTweets3 --username "CCHQPress" --since 2019-12-01 --until 2019-12-13
--output cchqpress.csv

GetOldTweets3 --username "Conservatives" --since 2019-12-01 --until
2019-12-13 --output cchqpress.csv
```







I also scraped the @Conservatives account as @CCHQPress is not their main account. This wasn't skewed in the same way (as seen on the left). This analysis is interesting as it could be demonstrative of a domination of more fear-driven or otherwise negative narratives from the Conservatives, as you tend to observe in

the right-wing media. Although after the successful result for Johson (et al.) it would be interesting to see how sentiment across right-wing media has varied. The scripts I wrote to do the sentiment analysis and graphing as well as all the csv files used throughout this work can be found here.

Python libraries used:

- Natural Language Toolkit
- Pandas
- GetOldTweets3
- Matplotlib

Summary

It has been interesting to briefly explore these networks, particularly using sentiment analysis which could be used to further explore the psychological composition of online political campaigns. More detailed analysis would be needed to have a more concrete conclusion to my original questions, however some interesting trends have been shown. This exploratory exercise has served as a good way of discovering some more specific research questions and areas of enquiry. A simple extension of this work would be to combine Sentiment analysis with the Gephi Graphs (using colour for example), so as to visualize the spread of sentiment in the networks. It would also be interesting to develop an automated online system, that does sentiment analysis on various key political accounts to see how it is spread and varies across party-political lines.